## Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Out of 40 students, 12 practise a given sport. Proportion?

2. If $P(A) = 0{,}3$, give $P(\overline{A})$.

3. Write "A and B" in set notation.

4. Express "A or B" with the inclusive or.

5. Calculate $0{,}4 \times 0{,}25$.

6. A fair die is rolled. Give $P(\{2; 4; 6\})$.

**After the diagnostic check.** Go over the flagged concepts with the teacher; redo a similar question, then explain the rule used.

**Oral work and logic.** Explain aloud what "given", "and" and "at least" mean. On a tree, justify an addition and a multiplication.

## Investigate before formalising

Build a table of 100 people in which A and B are independent, with 40 A and 30 B.
Change two cells while keeping these margins so as to break independence.

**Organisation.** Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min.
Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.